Python Engineering - Class 08

Data Processing | EX16 + EX16b - Masks and Filtering

Class objective: Learn to select, filter, summarize and validate numerical data with Boolean masks and NumPy operations.

This document is the base exercise guide. It is not a full solution to copy: it organizes the problem, implementation steps, reference values and required validation.

Path

```
01_python_bootcamp/
`-- week_02_data_structures/class_08_data_processing/
    |-- notes/
    `-- exercices/
```

Required workflow

```
PROBLEM
INPUTS
OUTPUTS
EQUATIONS
SPECIAL CASES
DO I NEED ITERATION?
ALGORITHM
VALIDATION
```

Before coding, write these eight sections. The goal is to separate problem reasoning from syntax.

EX16 - Trajectory Data Processing

Problem: Analyze an [h,v,t] matrix to identify ascent/descent, intervals and statistics.

Reference inputs: trajectory NumPy array

Equations / relationship: Boolean conditions + statistics + energy

Expected check: Count mask with np.sum; indices with np.where; new [h,v,t,energy] matrix; dt differences nearly constant.

Implementation guide

- 1. Create masks.
- 2. Combine conditions with & and |.
- 3. Filter rows.
- 4. Compute statistics.
- 5. Get indices.
- 6. Build new matrix.
- 7. Validate dt.

Before running

Predict sign, order of magnitude, data type and any relevant shape. Then compare the prediction with the actual output.

EX16b - Masks and Filtering Practice

Problem: Reinforce selection and reduction operations using smaller cases.

Reference inputs: reference arrays

Equations / relationship: Boolean filtering and reductions

Expected check: Understand x[mask] vs A[mask], np.min with axis, and negating conditions.

Implementation guide

- 1. Predict mask.
- 2. Apply it.
- 3. Compare values and indices.
- 4. Practice axis.
- 5. Explain each resulting shape.

Before running

Predict sign, order of magnitude, data type and any relevant shape. Then compare the prediction with the actual output.

Final validation

Answer these questions as comments at the end of the exercise. The goal is not memorized wording but demonstrating that you can explain the code flow.

- What is a Boolean mask?
- Why use & instead of and with arrays?
- Difference between A[mask] and np.where(mask).
- Why does np.diff return N-1 elements?
- What does np.allclose check?
- What does ~(v>0) mean?

Completion criterion

The class is complete when the code runs, the reference checks make sense, you can explain every block without reading a solution, and you can modify at least one input while predicting the expected effect.